

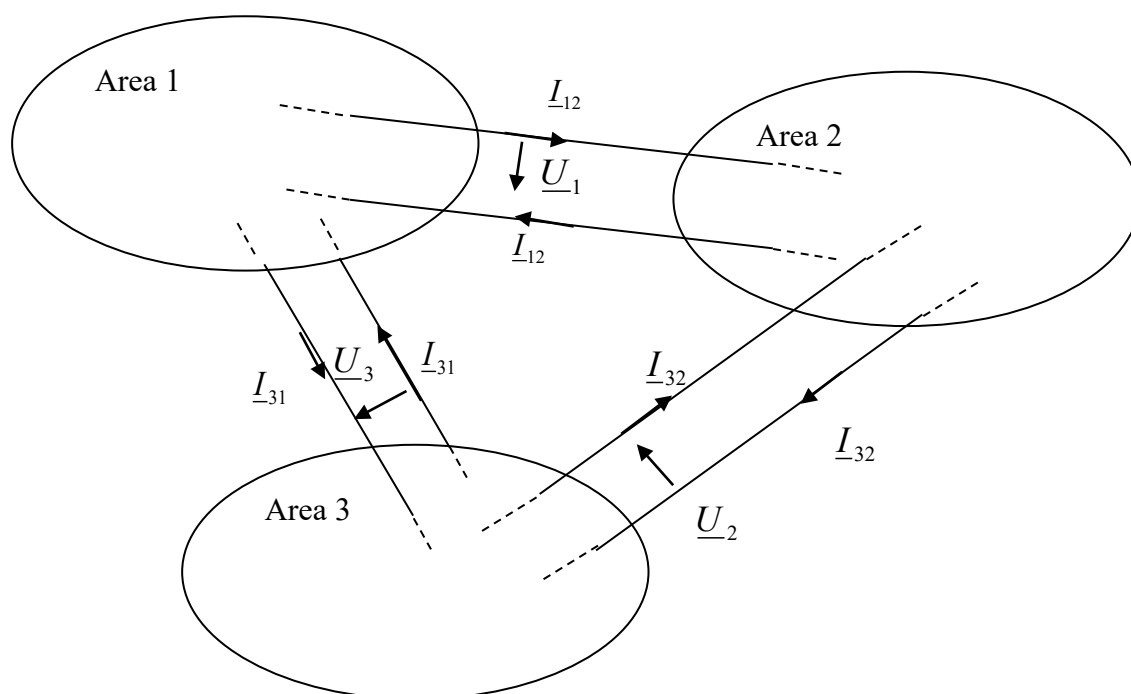
## Exercices Electric Power Grids (EPG)

### EPG-Exercise 1: Interpretation of numerical values in sign convention

Three areas are connected to each other by single-phase lines as shown in the figure below. The following simultaneous current and voltage values are measured:

Data:  $\underline{U}_1 = 76 \angle 4^\circ \text{ kV}$ ,  $\underline{U}_2 = 76 \angle 8^\circ \text{ kV}$ ,  $\underline{U}_3 = 76 \angle -11^\circ \text{ kV}$   
 $\underline{I}_{12} = 450 \angle -15^\circ \text{ A}$ ,  $\underline{I}_{31} = 300 \angle 18^\circ \text{ A}$ ,  $\underline{I}_{32} = 500 \angle -11^\circ \text{ A}$

1. Which areas provide active power? How much?
2. Which areas receive reactive power? How much?



$$S_1 = U_1 I_{12}^* - U_3 I_{31}^*$$

$$S_1 = 76 \angle 4^\circ \text{ kV} \cdot 450 \angle 15^\circ \text{ A} - 76 \angle -11^\circ \text{ kV} \cdot 300 \angle -18^\circ \text{ A}$$

$$= 34'200 \angle 19^\circ \text{ kW} - 22'800 \angle -29^\circ \text{ kW} \rightarrow \text{provider}$$

$$= 25'415,7 \angle 60,81^\circ \text{ kW} = 12'395,41 \text{ kVA} + j22'188,09 \text{ kVar}$$

$$S_2 = U_2 I_{32}^* - U_1 I_{12}^* = 76 \angle 8^\circ \text{ kV} \cdot 500 \angle 11^\circ \text{ A} - 76 \angle 4^\circ \text{ kV} \cdot 450 \angle 15^\circ \text{ A}$$

$$= 38'000 \angle 19^\circ \text{ kW} - 34'200 \angle 19^\circ \text{ kW} = 3'800 \angle 19^\circ \text{ kW}$$

$$= 3592,97 \text{ kVA} + j1237,16 \text{ kVar} \rightarrow \text{provider}$$

$$S_3 = U_3 I_{31}^* - U_2 I_{32}^* = 76 \angle -11^\circ \text{ kV} \cdot 300 \angle -18^\circ \text{ A} - 76 \angle 8^\circ \text{ kV} \cdot 500 \angle 11^\circ \text{ A}$$

$$= 22'800 \angle -29^\circ \text{ kW} - 38'000 \angle 19^\circ \text{ kW}$$

$$= 28'361,43 \angle -124,3^\circ \text{ kW} = -15'988,37 \text{ kVA} - j23'425,25 \text{ kVar}$$

↓  
receiver

$$1) \quad \underline{S} = \underline{U} \cdot \underline{I}^* = \underline{U} \cdot \underline{I} \angle -\theta$$

$$\begin{aligned} \underline{S}_1 &= \underline{U}_1 \underline{I}_{12}^* - \underline{U}_3 \underline{I}_{31}^* = 76 \angle 4^\circ \text{ kV} \cdot 450 \angle 15^\circ \text{ A} - 76 \angle -11^\circ \text{ kV} \cdot 300 \angle -18^\circ \text{ A} \\ &= 34'200 \angle 19^\circ \text{ kW} - 22'800 \angle 29^\circ \text{ kW} = 25'415,69^\circ \angle 60,81^\circ \text{ kW} \\ &= 12,4 + j22,2 \text{ MW} = 12,4 \text{ MW} + j22,2 \text{ MVar} \end{aligned}$$

$$\begin{aligned} \underline{S}_2 &= \underline{U}_2 \underline{I}_{32}^* - \underline{U}_1 \underline{I}_{12}^* = 76 \angle 8^\circ \text{ kV} \cdot 500 \angle 11^\circ \text{ A} - 76 \angle 4^\circ \text{ kV} \cdot 450 \angle 15^\circ \text{ A} \\ &= 38'000 \angle 19^\circ - 34'200 \angle 19^\circ \text{ kW} \\ &= 3,59 \text{ MW} + j1,24 \text{ MVar} \end{aligned}$$

$$\begin{aligned} \underline{S}_3 &= \underline{U}_3 \underline{I}_{31}^* - \underline{U}_2 \underline{I}_{32}^* = 76 \angle -11^\circ \text{ kV} \cdot 300 \angle -18^\circ \text{ A} - 76 \angle 8^\circ \text{ kV} \cdot 500 \angle 11^\circ \text{ A} \\ &= 22'800 \angle -29^\circ \text{ kW} - 38'000 \angle 19^\circ \text{ kW} \\ &= -15,99 \text{ MW} - j23,43 \text{ MVar} \end{aligned}$$

$P_1, Q_1, P_2, Q_2$  : Provider

$P_3, Q_3$  : receiver

## EPG-Exercise 2: Merit order principle for pricing on the spot market

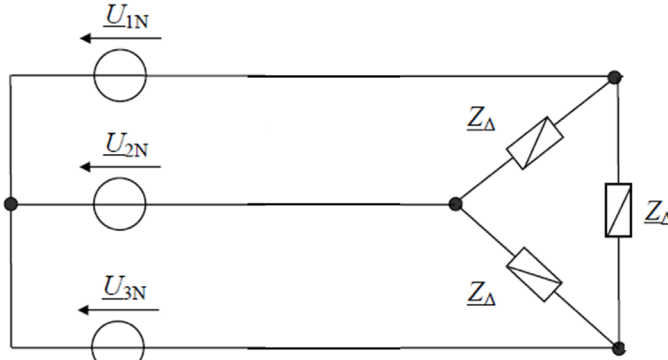
The energy prices are formed on the spot market with the help of a double auction. Sorted list of B1-B6 consumer offers:

| B1           | B2           | B3           | B4           | B5           | B6           |
|--------------|--------------|--------------|--------------|--------------|--------------|
| 100 MWh      | 200 MWh      | 250 MWh      | 250 MWh      | 150 MWh      | 150 MWh      |
| 0.08 Fr./kWh | 0.09 Fr./kWh | 0.10 Fr./kWh | 0.11 Fr./kWh | 0.12 Fr./kWh | 0.13 Fr./kWh |

Sorted list of offers of electricity sellers S1-S5:

| S1           | S2            | S3            | S4            | S5           |
|--------------|---------------|---------------|---------------|--------------|
| 50 MWh       | 200 MWh       | 250 MWh       | 250 MWh       | 400 MWh      |
| 0.07 Fr./kWh | 0.075 Fr./kWh | 0.085 Fr./kWh | 0.095 Fr./kWh | 0.15 Fr./kWh |

## EPG-Exercise 3: Three-phase power



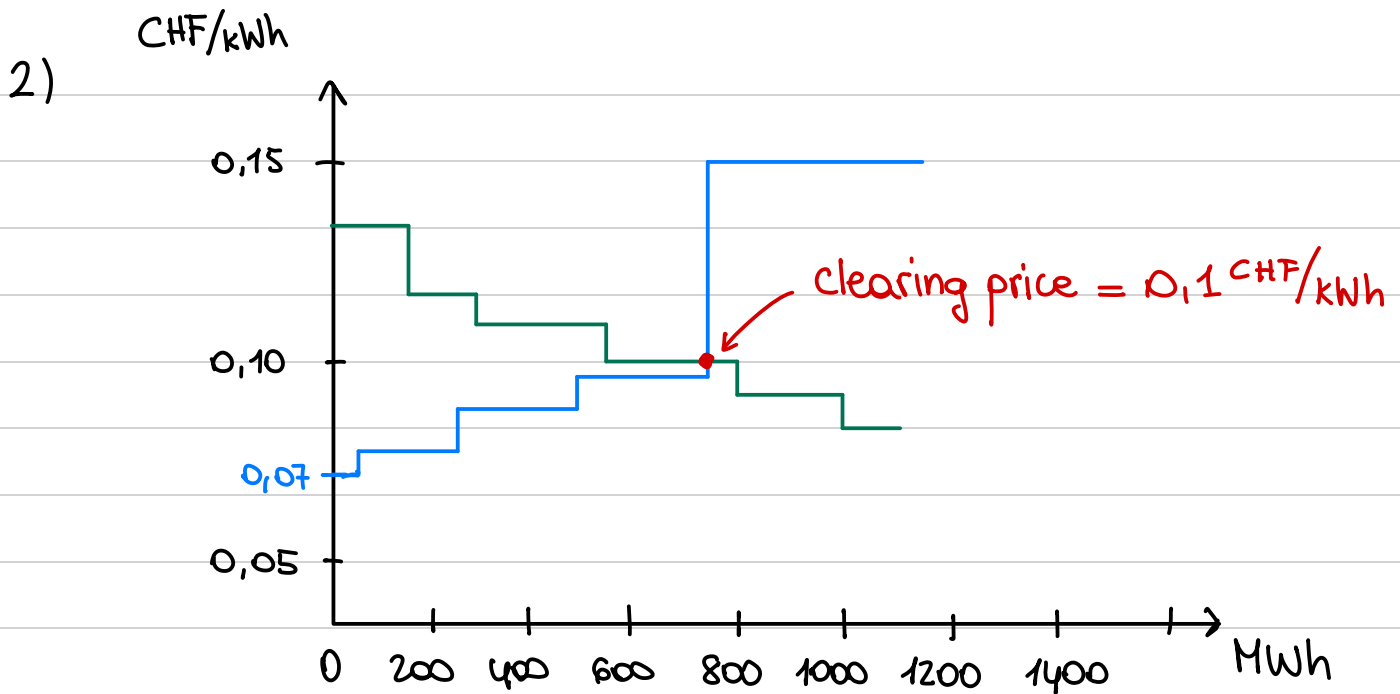
$$U = RI$$

$$\underline{U} = \underline{Z} \underline{I} \quad \underline{I} = \frac{\underline{U}_\Delta}{\underline{Z}}$$

Data

|                                                     |                                                        |                                                       |
|-----------------------------------------------------|--------------------------------------------------------|-------------------------------------------------------|
| $\underline{U}_{1N} = 110 \text{ V} \angle 0^\circ$ | $\underline{U}_{2N} = 230 \text{ V} \angle -120^\circ$ | $\underline{U}_{3N} = 230 \text{ V} \angle 120^\circ$ |
| $\underline{Z}_\Delta = (30 + j25) \Omega$          |                                                        |                                                       |

Calculate the complex power  $\underline{S}$  and the power factor  $\underline{\lambda}$ .



$$3) \quad \underline{U}_{12} = \underline{U}_{1N} - \underline{U}_{2N} = 110 \text{ V} - 230 \text{ V} \angle -120^\circ = 300,5 \angle 41,5^\circ \text{ V}$$

$$\underline{U}_{23} = \underline{U}_{2N} - \underline{U}_{3N} = 230 \angle -120^\circ \text{ V} - 230 \angle 120^\circ \text{ V} = 398,37 \angle -90^\circ \text{ V}$$

$$\underline{U}_{31} = \underline{U}_{3N} - \underline{U}_{1N} = 230 \angle 120^\circ \text{ V} - 110 \text{ V} = 300,5 \angle 138,5^\circ \text{ V}$$

$$\underline{I}_{12} = \frac{\underline{U}_{12}}{\underline{Z}_\Delta} = 7,695 \angle 1,7^\circ \text{ A}$$

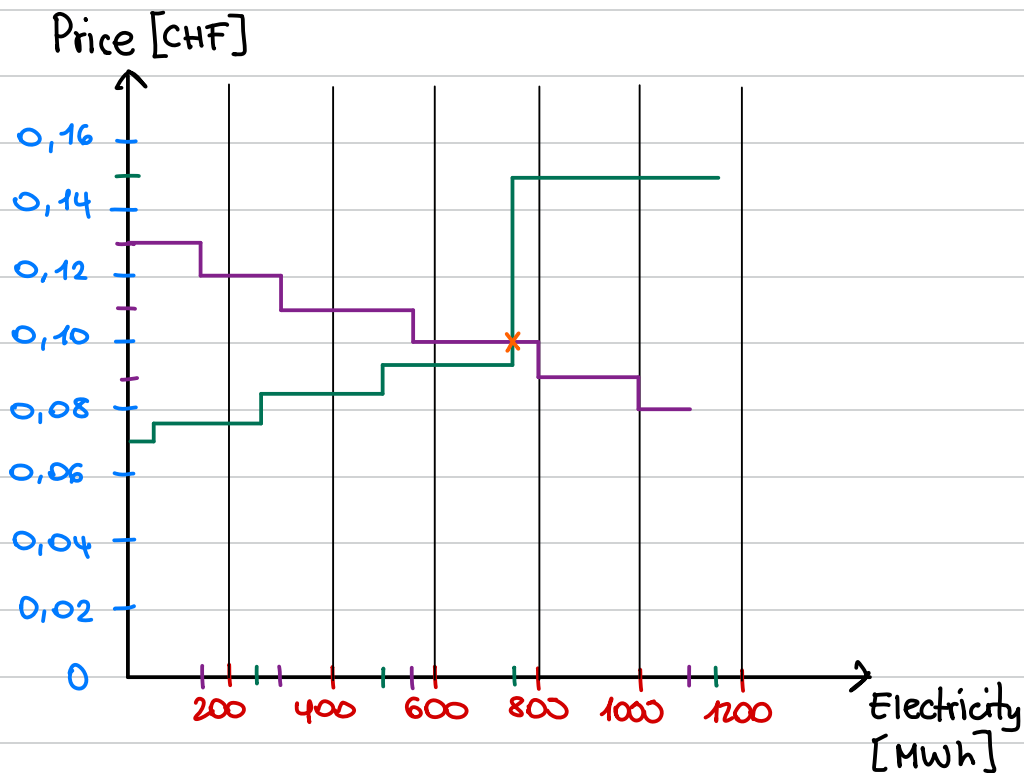
$$\underline{I}_{23} = \frac{\underline{U}_{23}}{\underline{Z}_\Delta} = 10,2 \angle -129,8^\circ \text{ A}$$

$$\underline{I}_{31} = \frac{\underline{U}_{31}}{\underline{Z}_\Delta} = 7,695 \angle 98,7^\circ \text{ A}$$

$$\begin{aligned} S &= \underline{U}_{12} \cdot \underline{I}_{12}^* + \underline{U}_{23} \cdot \underline{I}_{23}^* + \underline{U}_{31} \cdot \underline{I}_{31}^* = 6'674,9 \text{ kVA} + j \cdot 5'561,3 \text{ kVar} \\ &= 8'688,1 \angle 39,8^\circ \text{ kW} \end{aligned}$$

$$\lambda = \cos(39,8^\circ) = 0,768$$

2)



x → clearing price = 0,1 CHF

3)

$$S = \frac{|\underline{U}^2|}{\underline{Z}_{\Delta}^*}$$

$$\underline{S} = \frac{|\underline{U}_{12}|^2 + |\underline{U}_{23}|^2 + |\underline{U}_{31}|^2}{\underline{Z}_{\Delta}^*}$$

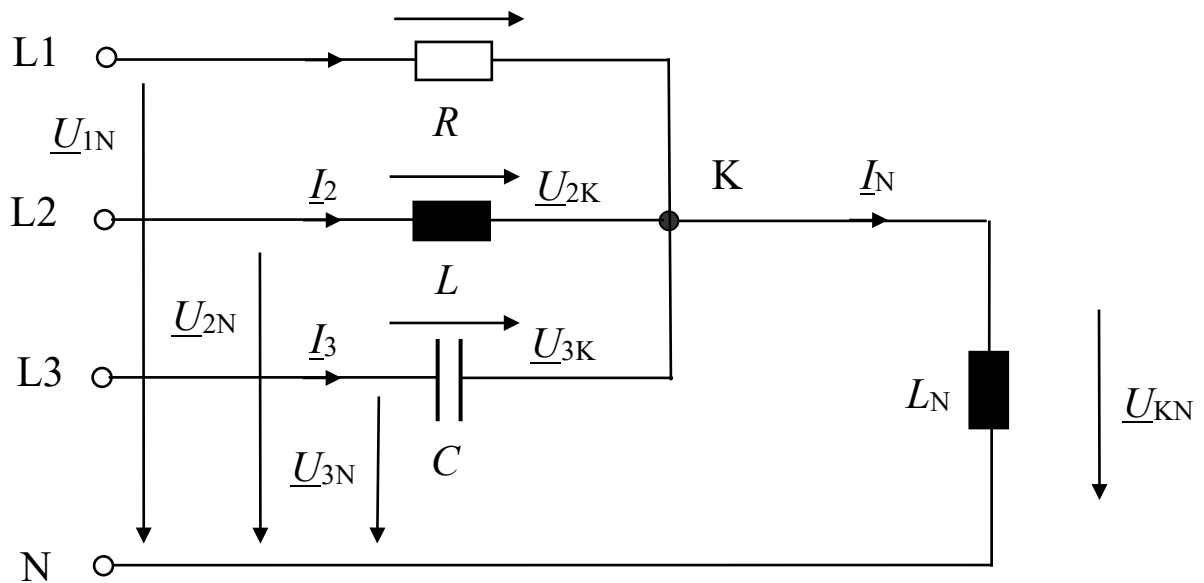
$$\underline{U}_{12} = \underline{U}_{1N} - \underline{U}_{2N} \quad \underline{U}_{12} = 110 \angle 0^\circ \text{ V} - 230 \angle -120^\circ \text{ V} = 300,5 \angle 41,52^\circ \text{ V}$$

$$\underline{U}_{23} = \underline{U}_{2N} - \underline{U}_{3N} \Rightarrow \underline{U}_{23} = 230 \angle -120^\circ \text{ V} - 230 \angle 120^\circ \text{ V} = 398,5 \angle -90^\circ \text{ V}$$

$$\underline{U}_{31} = \underline{U}_{3N} - \underline{U}_{1N} \quad \underline{U}_{31} = 230 \angle 120^\circ \text{ V} - 110 \angle 0^\circ \text{ V} = 300,5 \angle 138,48^\circ \text{ V}$$

$$\underline{S} = \frac{300,5^2 \text{ V} + 300,5^2 \text{ V} + 398,5^2 \text{ V}}{39,05 \angle -39,81^\circ \Omega} = 6,68 + j5,56 \text{ kVA} = 8,69 \angle 39,81^\circ \text{ kW}$$

$$\lambda = \cos(39,81^\circ) = 0,768$$

**EPG-Exercise 4: 4-wire system**

|       |                                                         |                         |
|-------|---------------------------------------------------------|-------------------------|
| Data: | $\underline{U}_{1N} = 220 \text{ V } \angle 0^\circ$    | $R = 100 \, \Omega$     |
|       | $\underline{U}_{2N} = 220 \text{ V } \angle -120^\circ$ | $L = 160 \text{ mH}$    |
|       | $\underline{U}_{3N} = 220 \text{ V } \angle 120^\circ$  | $C = 40 \, \mu\text{F}$ |
|       | $f = 50 \text{ Hz}$                                     | $L_N = 500 \text{ mH}$  |

Calculate the currents  $\underline{I}_1$ ,  $\underline{I}_2$ ,  $\underline{I}_3$  and  $\underline{I}_N$ , as well as the voltages  $\underline{U}_{1K}$ ,  $\underline{U}_{2K}$ ,  $\underline{U}_{3K}$  and  $\underline{U}_{KN}$ .

$$4) \underline{U}_{KN} = \frac{\underline{Y}_1 \underline{U}_{1N} + \underline{Y}_2 \underline{U}_{2N} + \underline{Y}_3 \underline{U}_{3N}}{\underline{Y}_1 + \underline{Y}_2 + \underline{Y}_3 + \underline{Y}_N}$$

$$\underline{Y}_1 = \frac{1}{R} = 0,01 \text{ } 1/\Omega$$

$$\underline{Y}_2 = \frac{1}{2\pi f L} \angle -90^\circ = \frac{1 \angle -90^\circ}{2\pi \cdot 50 \text{ Hz} \cdot 160 \cdot 10^{-3} \text{ H}} = 0,0199 \angle -90^\circ \text{ } 1/\Omega$$

$$\underline{Y}_3 = 2\pi f C \angle 90^\circ = 2\pi \cdot 50 \text{ Hz} \cdot 40 \cdot 10^{-6} \text{ F} \angle 90^\circ = 0,01257 \angle 90^\circ \text{ } 1/\Omega$$

$$\underline{Y}_N = \frac{1}{2\pi f L} \angle -90^\circ = \frac{1 \angle -90^\circ}{2\pi \cdot 50 \text{ Hz} \cdot 500 \cdot 10^{-3} \text{ H}} = 0,00637 \angle -90^\circ \text{ } 1/\Omega$$

$$\underline{U}_{KN} = \frac{4,0671 \angle 168,565^\circ \text{ V}/\Omega}{0,01696 \angle -53,87^\circ \text{ } 1/\Omega} = 239,785 \angle -137,56^\circ \text{ V}$$



$$4) \quad \underline{U}_{KN} = \frac{\underline{Y}_1 \underline{U}_{1N} + \underline{Y}_2 \underline{U}_{2N} + \underline{Y}_3 \underline{U}_{3N}}{\underline{Y}_1 + \underline{Y}_2 + \underline{Y}_3 + \underline{Y}_N}$$

$$\underline{Y}_1 = \frac{1}{R} \angle 0^\circ = 0,01 \Omega$$

$$\underline{Y}_2 = \frac{1}{\omega L} \angle -90^\circ = \frac{1}{2\pi \cdot 50 \text{ Hz} \cdot 0,16 \text{ H}} \angle -90^\circ = 0,0199 \angle -90^\circ \Omega$$

$$\underline{Y}_3 = \omega C \angle 90^\circ = 2\pi \cdot 50 \text{ Hz} \cdot 40 \cdot 10^{-6} \text{ F} \angle 90^\circ = 0,0126 \angle 90^\circ \Omega$$

$$\underline{Y}_N = \frac{1}{\omega L_N} \angle -90^\circ = \frac{1}{2\pi \cdot 50 \text{ Hz} \cdot 0,5 \text{ H}} \angle -90^\circ = 0,00637 \angle -90^\circ \Omega$$

$$\begin{aligned} \underline{U}_{KN} &= \frac{0,01 \Omega \cdot 220 \text{ V} + 0,02 \angle -90^\circ \Omega \cdot 220 \angle -120^\circ + 0,0125 \angle 90^\circ \Omega \cdot 220 \angle 120^\circ \text{ V}}{0,01 \Omega + 0,02 \angle -90^\circ \Omega + 0,0125 \angle 90^\circ \Omega + 0,00637 \angle -90^\circ \Omega} \\ &= \frac{2,2 + 4,4 \angle 150^\circ + 2,75 \angle -150^\circ}{0,0171 \angle -54,21^\circ} = \frac{4,076 \angle 168,32^\circ \text{ V} \cdot \Omega}{0,0171 \angle -54,21^\circ \Omega} = \underline{238,4 \angle -137,5^\circ \text{ V}} \end{aligned}$$

$$\underline{U}_{1K} = \underline{U}_{1N} - \underline{U}_{KN} = 220 \text{ V} - 238,4 \angle -137,5^\circ \text{ V} = \underline{427,2 \angle 22^\circ \text{ V}}$$

$$\underline{U}_{2K} = \underline{U}_{2N} - \underline{U}_{KN} = 220 \angle -120^\circ - 238,4 \angle -137,5^\circ \text{ V} = \underline{71,9 \angle -24^\circ \text{ V}}$$

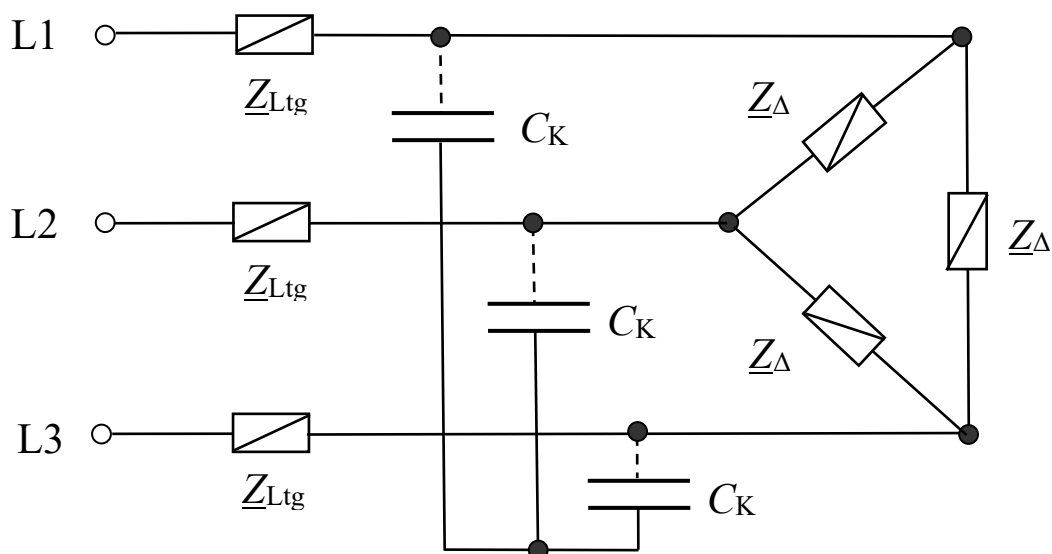
$$\underline{U}_{3K} = \underline{U}_{3N} - \underline{U}_{KN} = 220 \angle 120^\circ - 238,4 \angle -137,5^\circ \text{ V} = \underline{357,8 \angle 79,4^\circ \text{ V}}$$

$$\underline{I}_1 = \underline{U}_{1K} \cdot \underline{Y}_1 = 427,2 \angle 22^\circ \text{ V} \cdot 0,01 \angle 0^\circ = \underline{4,27 \angle 22^\circ \text{ A}}$$

$$\underline{I}_2 = \underline{U}_{2K} \cdot \underline{Y}_2 = 71,9 \angle -24^\circ \text{ V} \cdot 0,0199 \angle -90^\circ \Omega = \underline{1,43 \angle -114^\circ \text{ A}}$$

$$\underline{I}_3 = \underline{U}_{3K} \cdot \underline{Y}_3 = 357,8 \angle 79,4^\circ \text{ V} \cdot 0,0126 \angle 90^\circ \Omega = \underline{4,51 \angle 169,4^\circ \text{ A}}$$

$$\underline{I}_N = \underline{U}_{KN} \cdot \underline{Y}_N = 238,4 \angle -137,5^\circ \text{ V} \cdot 0,00637 \angle -90^\circ \Omega = \underline{1,52 \angle 132,5^\circ \text{ A}}$$

**EPG-Exercise 5: Calculation of Three-Phase Systems**

Data:

$$U = 400 \text{ V} / 50 \text{ Hz (line-to-line voltage)}$$

$$\underline{Z}_{Ltg} = (1 + j \cdot 2) \, \Omega$$

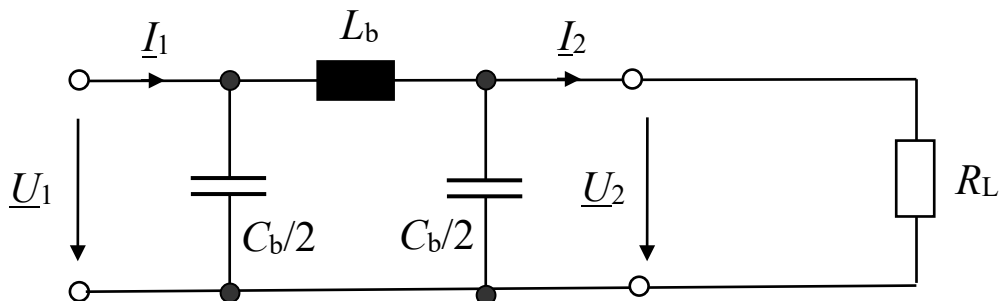
$$\underline{Z}_{\Delta} = 30 \angle 60^\circ \, \Omega$$

- Calculate the line currents  $\underline{I}$  and voltages  $\underline{U}_{\Delta}$  and the loads  $\underline{Z}_{\Delta}$ , if the capacitor bank is not connected.
- Calculate the value of  $C_K$  in the capacitor bank for reactive current compensation of the consumers  $\underline{Z}_{\Delta}$ .
- Determine the line currents  $\underline{I}$  in the compensated case.



### EPG-Exercise 6: Surge impedance loading

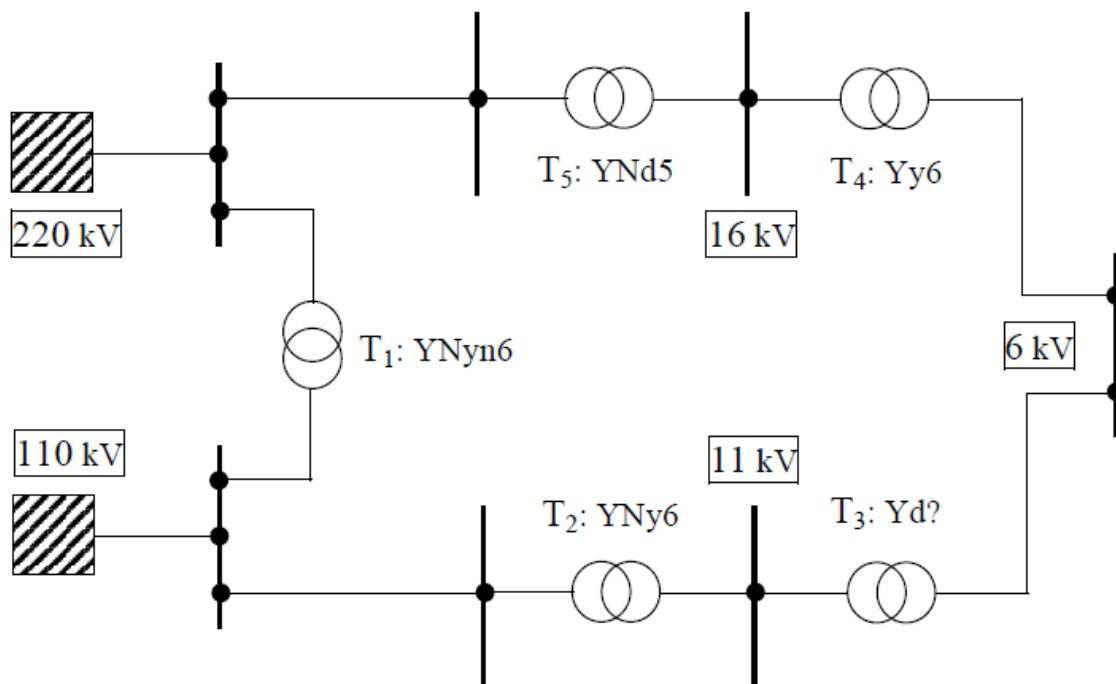
Equivalent line model of a lossless line with load  $R_L$ :



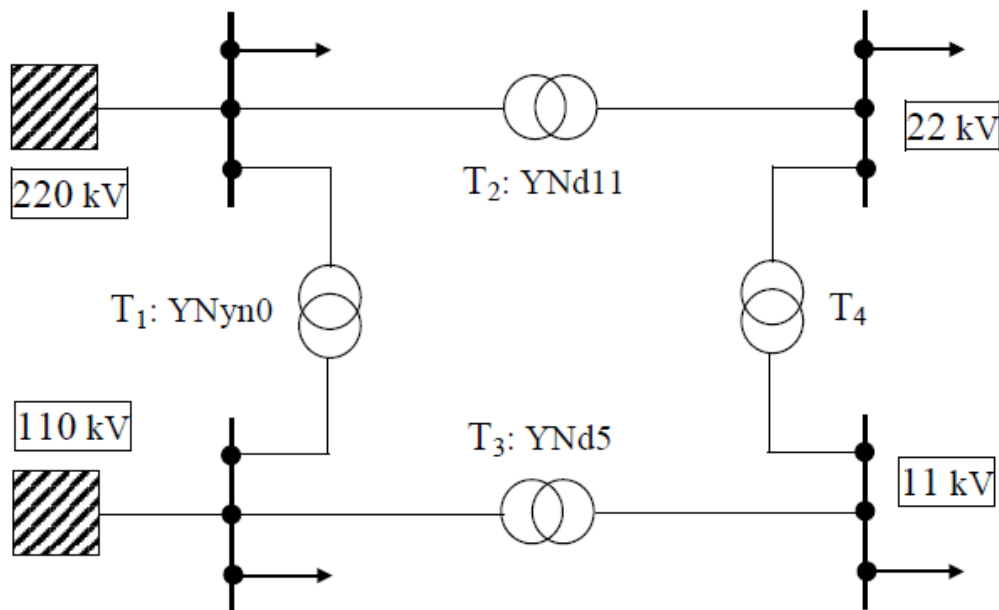
Determine the value of  $R_L$  so that the lossless line is in optimal operation (i.e. surge impedance loading) with:

$$\begin{aligned} L_b' &= 1.3 \text{ mH/(km and phase)} \\ C_b' &= 9.6 \text{ nF/(km and phase)} \\ l &= 250 \text{ km (length of the line)} \end{aligned}$$

### EPG-Exercise 7: Transformer 1



Determine the index number of transformer T3.

**EPG-Exercise 8: Transformer 2**

Determine a suitable transformer configuration incl. index number for transformer T4.

**EPG-Exercise 9: Active and reactive power of a synchronous generator**

A three-phase synchronous generator (star connected) is built as a nonsalient pole machine. It operates in a grid with constant voltage and frequency (stiff grid).

Data:

Nominal apparent power:  $S_N = 400 \text{ MVA}$

Nominal voltage\*:  $U_N = 20 \text{ kV} / 50 \text{ Hz}$

Synchronous reactance:  $X_d = 2 \text{ p.u.}$

In the operation considered, the excitation current is set so that the induced rotor voltage reaches 25 kV (phase voltage). The torque angle is  $30^\circ$ . Determine the active and reactive power of the generator.

What is the electrical behavior of the machine? (capacitive or inductive?)

**\*Important: if nothing specified, nominal voltage is always the line-to-line voltage.**

$$g) \quad P = \frac{U \cdot U_P}{X_d} \sin(\vartheta) = \frac{20 \cdot 10^3 V \cdot 25 \cdot 10^3 V}{\sqrt{3} \cdot 2 \text{ p.u.}} \sin(30^\circ) = 72,2 \text{ MW}$$

$$P_{3p} = 3 \cdot P = 72,2 \text{ MW} \cdot 3 = 216,6 \text{ MW}$$

$$Q = \frac{U \cdot U_P}{X_d} \cos(\vartheta) - \frac{U^2}{X_d} = \frac{20 \cdot 10^3 V \cdot 25 \cdot 10^3 V}{\sqrt{3} \cdot 2 \text{ p.u.}} \cos(30^\circ) - \frac{\left(\frac{20 \cdot 10^3 V}{\sqrt{3}}\right)^2}{2 \text{ p.u.}}$$

$$= 58,33 \text{ MVA}$$

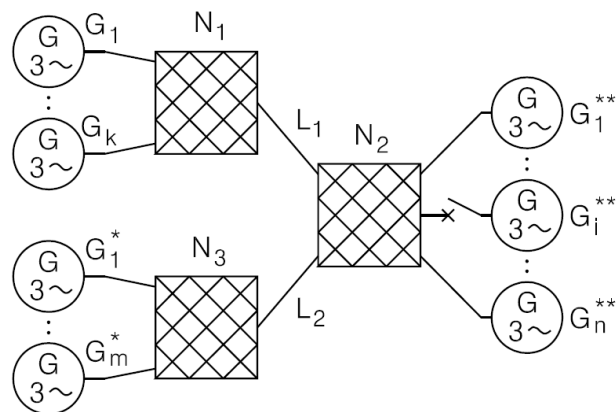
$$Q_{3p} = 3 \cdot Q = 58,33 \text{ MVA} \cdot 3 = 175 \text{ MVA}$$

### EPG-Exercise 10: Interconnected control areas

The interconnected control areas of the grid below are examined ( $N_1$ ,  $N_2$ , and  $N_3$  could, e.g., be three transmission grids of three countries, that are interconnected through the lines  $L_1$  and  $L_2$ , respectively).

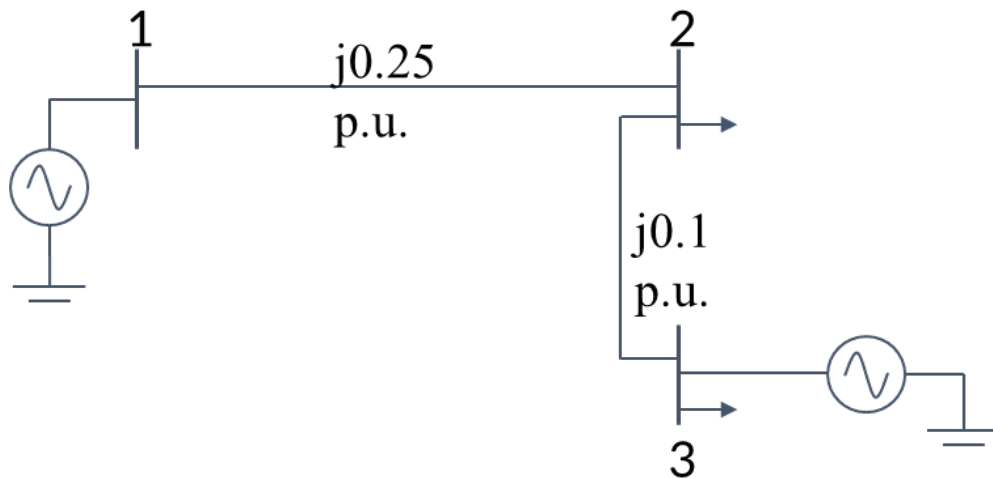
At the considered time no power is exchanged on the lines  $L_1$  and  $L_2$ . Due to a fault, a power plant fails in the control area  $N_2$ . This plant has previously fed 100 MW of power into the grid. The three control areas have the regulating characteristics  $K_{N1} = 400 \text{ MW/Hz}$  and  $K_{N2} = K_{N3} = 500 \text{ MW/Hz}$ .

- What frequency will occur in the control areas after the primary controllers have been activated?
- What power is then exchanged between the areas?
- What input signals  $\Delta P_R$  are given to the secondary controllers of the three areas?
- Approximately after which time period will the power exchange on the grid lines  $L_1$  and  $L_2$  be back to normal (i.e., before the disturbance assuming a constant load)?



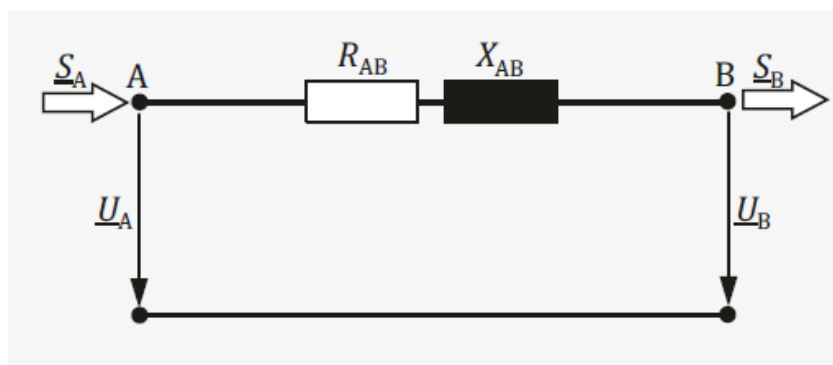
### ✓ EPG-Exercise 11: Ybus

Form the Ybus of the following system:



### ✓ EPG-Exercise 12: Two-node system

For the following 2-node system, program a simple iterative load flow calculation (e.g. Excel, Matlab) to calculate  $\underline{U}_B$ :



Given:

$$U_A = |U_A| = 400V$$

$$Z_{AB} = 0.1 + j0.1 \Omega$$

$$P_B = 9815 W$$

$$Q_B = 4908 Var$$

To be calculated:

$\underline{U}_B$  (magnitude and phase angle)

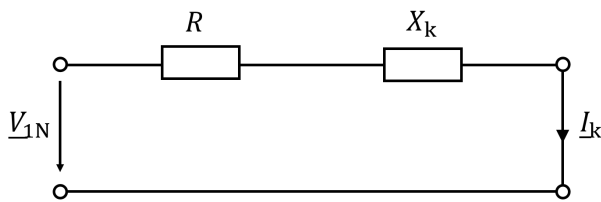


$$11) \begin{cases} I_1 = \frac{1}{j0,25} (v_1 - v_2) \\ I_2 = \frac{1}{j0,1} (v_2 - v_3) + (v_2 - v_1) \cdot \frac{1}{j0,25} \\ I_3 = \frac{1}{j0,1} (v_3 - v_2) \end{cases}$$

$$\begin{cases} I_1 = -j4 V_1 + j4 V_2 \\ I_2 = -j10 V_2 + j10 V_3 - j4 V_2 + j4 V_1 \\ I_3 = -j10 V_3 + j10 V_2 \end{cases} \Rightarrow \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} -j4 & j4 & 0 \\ j4 & -j14 & j10 \\ 0 & j10 & -j10 \end{bmatrix} \cdot \vec{V}$$

$$12) \text{ Using python: } \underline{V_B} = 396,3 - 1,23j \text{ V} = 396,3 \angle -0,18^\circ \text{ V}$$

### ✓ EPG-Exercise 13: Short circuits



$$V = 20\text{kV (line voltage)}$$

$$f = 50\text{Hz}$$

$$I_k = 300\text{ A (Steady state short circuit current)}$$

$$R = 12\Omega$$

- Determine the value of the leakage reactance  $X_k$ .
- Calculate the short-circuit current at time  $t = 0.03\text{s}$ . Assume that the short circuit occurred at  $t = 0$ , i.e., when the voltage crossed zero (rising edge).

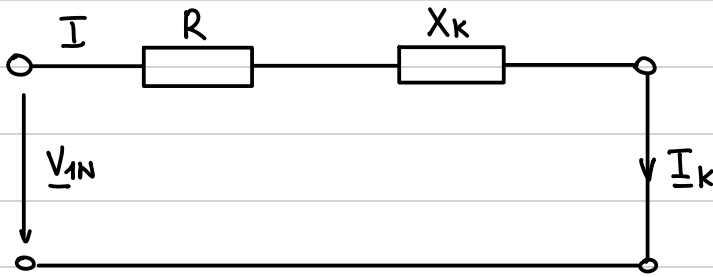
$$i(t) = i_s(t) + i_t(t)$$

$$i_s(t) = \frac{\sqrt{2}V_s}{|Z|} \sin(\omega t + \alpha - \theta)$$

$$i_t(t) = -i_s(0) \cdot e^{-(R/L)t} = \frac{\sqrt{2}V_s}{|Z|} \sin(\theta - \alpha) \cdot e^{-(R/L)t}$$

$$\text{with } |Z| = \sqrt{R^2 + \omega^2 L^2} \angle \left( \theta = \tan^{-1} \frac{\omega L}{R} \right)$$

## Short circuit



$$a) \quad V_{1N} = \frac{V_{line}}{\sqrt{3}}$$

$$I_k = \frac{V_{1N}}{|z|} \Rightarrow |z| = \frac{V_{line}}{\sqrt{3} I_k}$$

$$|z| = \sqrt{R^2 + X_k^2} \Rightarrow |z|^2 = R^2 + X_k^2 \Rightarrow X_k = \sqrt{|z|^2 - R^2} = \underline{36,57 \Omega}$$

$$b) \quad i_t(t) = -i_s(0) e^{-(R/k)t}$$

$$X_k = 2\pi f K \Rightarrow K = \frac{X_k}{2\pi f} = \frac{36,57 \Omega}{2\pi \cdot 50 \text{ Hz}} = 0,1164 \text{ H}$$

$$i_t(0,03 \text{ s}) = \sqrt{2} I_k \sin(\theta) \cdot e^{-(12/0,1164) \cdot 0,03}$$

$$= \sqrt{2} I_k \cdot \sin\left(\arctan\left(\frac{X_k}{R}\right)\right) e^{-(12/0,1164) \cdot 0,03} = 18,29 \text{ A}$$

$$i_s(0,03 \text{ s}) = \sqrt{2} I_k \cdot \sin\left(2\pi f \cdot 0,03 \text{ s} - \arctan\left(\frac{X_k}{R}\right)\right)$$

$$= 403,12 \text{ A}$$

$$i(t) = i_t(t) + i_s(t) = 403,12 \text{ A} + 18,29 \text{ A} = \underline{421,41 \text{ A}}$$